Set 3: Apparent and true dip

Answer key · CratonLab Pack 1 · Set 3 · 12 problems

Key: 1D 2D 3A 4B 5D 6B 7D 8A 9B 10C 11A 12C

1. **APPARENT DIP**
   A cross section runs along 130°–310° across plane **050/70**. What is the apparent dip in the section?

   Answer: (D) Apparent dip 69.7°

   1. Angle between strike (050°) and the section line: 80°.
   2. tan(apparent) = tan(70°) × sin(80°) → **69.7°**, dipping toward 130°.
   3. On the net: the point where the great circle crosses the 130° radius; count its plunge.

2. **TRUE DIP FROM TWO APPARENT DIPS**
   In two road cuts, a bed shows an apparent dip of **34.4° toward 200°** and **47.2° toward 170°**. Find the strike (RHR) and true dip.

   Answer: (D) Strike (RHR) 055°, Dip 50°

   1. Each apparent dip is a line lying in the bed: plot 34.4→200 and 47.2→170 as points.
   2. Rotate the overlay until both points lie on one great circle and trace it: that great circle is the bed.
   3. Read its strike and dip: **055/50**. The true dip is always steeper than any apparent dip.

3. **APPARENT DIP**
   A cross section runs along 155°–335° across plane **260/35**. What is the apparent dip in the section?

   Answer: (A) Apparent dip 34.1°

   1. Angle between strike (260°) and the section line: 75°.
   2. tan(apparent) = tan(35°) × sin(75°) → **34.1°**, dipping toward 335°.
   3. On the net: the point where the great circle crosses the 335° radius; count its plunge.

4. **TRUE DIP FROM TWO APPARENT DIPS**
   In two road cuts, a bed shows an apparent dip of **34.5° toward 305°** and **32.7° toward 020°**. Find the strike (RHR) and true dip.

   Answer: (B) Strike (RHR) 250°, Dip 40°

   1. Each apparent dip is a line lying in the bed: plot 34.5→305 and 32.7→020 as points.
   2. Rotate the overlay until both points lie on one great circle and trace it: that great circle is the bed.
   3. Read its strike and dip: **250/40**. The true dip is always steeper than any apparent dip.

5. **APPARENT DIP**
   A cross section runs along 255°–435° across plane **300/65**. What is the apparent dip in the section?

   Answer: (D) Apparent dip 56.6°

   1. Angle between strike (300°) and the section line: 45°.
   2. tan(apparent) = tan(65°) × sin(45°) → **56.6°**, dipping toward 075°.
   3. On the net: the point where the great circle crosses the 075° radius; count its plunge.

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6. **TRUE DIP FROM TWO APPARENT DIPS**
In two road cuts, a bed shows an apparent dip of **30.8° toward 015°** and **48.2° toward 335°**. Find the strike (RHR) and true dip.

Answer: (B) Strike (RHR) 225°, Dip 50°

1. Each apparent dip is a line lying in the bed: plot 30.8→015 and 48.2→335 as points.
2. Rotate the overlay until both points lie on one great circle and trace it: that great circle is the bed.
3. Read its strike and dip: **225/50**. The true dip is always steeper than any apparent dip.

7. **APPARENT DIP**
A cross section runs along 215°–395° across plane **300/30**. What is the apparent dip in the section?

Answer: (D) Apparent dip 29.9°

1. Angle between strike (300°) and the section line: 85°.
2. tan(apparent) = tan(30°) × sin(85°) → **29.9°**, dipping toward 035°.
3. On the net: the point where the great circle crosses the 035° radius; count its plunge.

8. **TRUE DIP FROM TWO APPARENT DIPS**
In two road cuts, a bed shows an apparent dip of **36.3° toward 295°** and **50.9° toward 170°**. Find the strike (RHR) and true dip.

Answer: (A) Strike (RHR) 135°, Dip 65°

1. Each apparent dip is a line lying in the bed: plot 36.3→295 and 50.9→170 as points.
2. Rotate the overlay until both points lie on one great circle and trace it: that great circle is the bed.
3. Read its strike and dip: **135/65**. The true dip is always steeper than any apparent dip.

9. **APPARENT DIP**
A cross section runs along 190°–370° across plane **355/65**. What is the apparent dip in the section?

Answer: (B) Apparent dip 29°

1. Angle between strike (355°) and the section line: 15°.
2. tan(apparent) = tan(65°) × sin(15°) → **29°**, dipping toward 010°.
3. On the net: the point where the great circle crosses the 010° radius; count its plunge.

10. **TRUE DIP FROM TWO APPARENT DIPS**
In two road cuts, a bed shows an apparent dip of **52.3° toward 160°** and **53.3° toward 205°**. Find the strike (RHR) and true dip.

Answer: (C) Strike (RHR) 095°, Dip 55°

1. Each apparent dip is a line lying in the bed: plot 52.3→160 and 53.3→205 as points.
2. Rotate the overlay until both points lie on one great circle and trace it: that great circle is the bed.
3. Read its strike and dip: **095/55**. The true dip is always steeper than any apparent dip.

11. **APPARENT DIP**
A cross section runs along 055°–235° across plane **155/35**. What is the apparent dip in the section?

Answer: (A) Apparent dip 34.6°

1. Angle between strike (155°) and the section line: 80°.
2. tan(apparent) = tan(35°) × sin(80°) → **34.6°**, dipping toward 235°.
3. On the net: the point where the great circle crosses the 235° radius; count its plunge.

12. **TRUE DIP FROM TWO APPARENT DIPS**

In two road cuts, a bed shows an apparent dip of **12.4° toward 180°** and **5.2° toward 215°**. Find the strike (RHR) and true dip.

Answer: (C) Strike (RHR) 055°, Dip 15°

1. Each apparent dip is a line lying in the bed: plot 12.4 → 180 and 5.2 → 215 as points.
2. Rotate the overlay until both points lie on one great circle and trace it: that great circle is the bed.
3. Read its strike and dip: **055/15**. The true dip is always steeper than any apparent dip.

Methods and data: Athabasca Univ. GEOL 319 Lab 3 (https://geol319r4.athabascau.ca/lab-unit03.htm); U. Alberta EAS 233 Lab 3 (https://courses.eas.ualberta.ca/eas233/0809winter/EAS233Lab03notes.pdf).